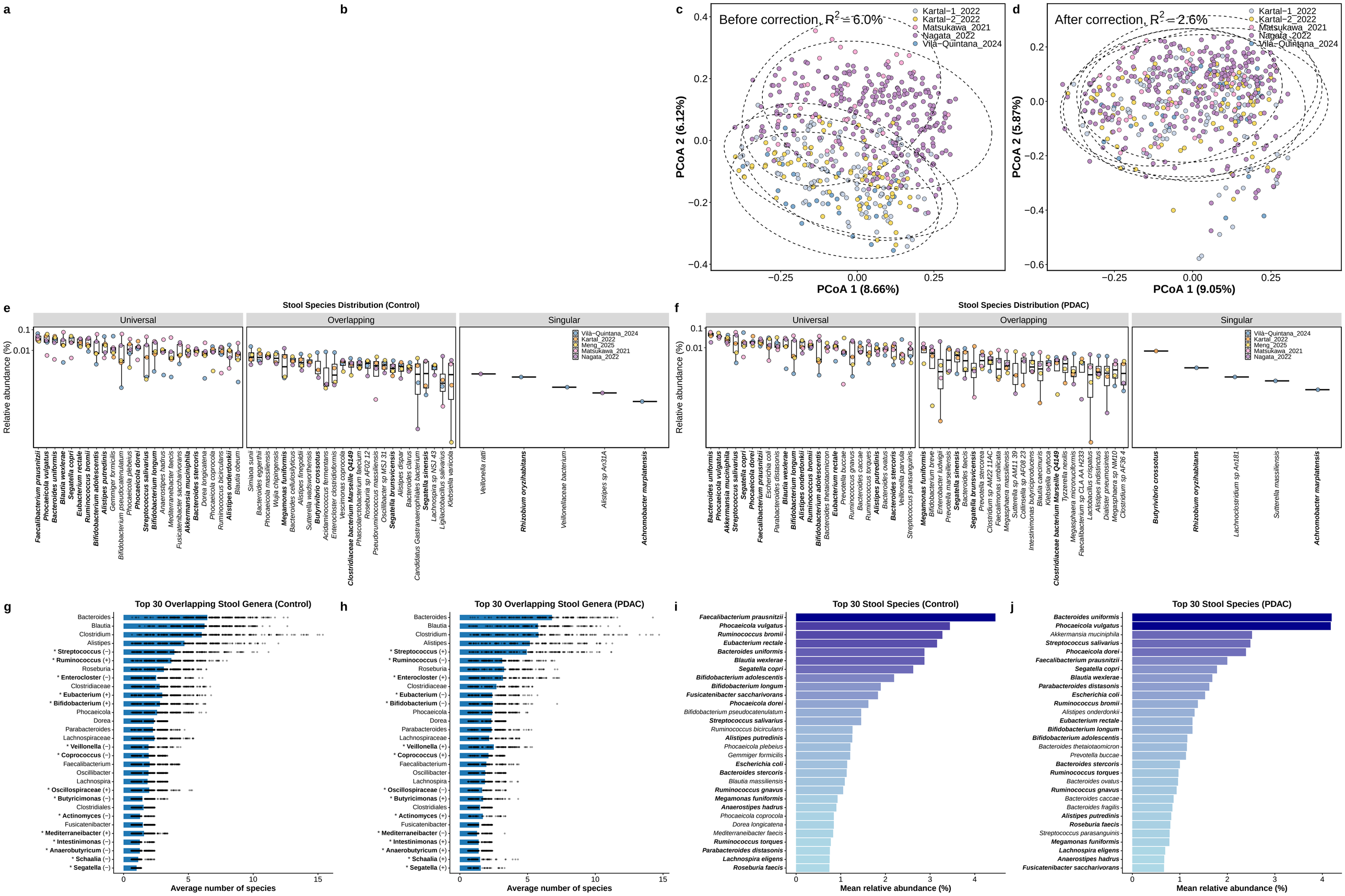

a
b
c
Before correction, R2 = 6.0%
Kartal−1_2022
Kartal−2_2022
Matsukawa_2021
Nagata_2022
Vilà−Quintana_2024
PCoA 2 (6.12%)
PCoA 1 (8.66%)
d
After correction, R2 = 2.6%
PCoA 2 (5.87%)
PCoA 1 (9.05%)
e
Stool Species Distribution (Control)
Universal
Overlapping
Singular
Relative abundance (%)
Vilà−Quintana_2024
Kartal_2022
Meng_2025
Matsukawa_2021
Nagata_2022
Faecalibacterium prausnitzii
Phocaeicola vulgatus
Bacteroides uniformis
Blautia wexlerae
Segatella copri
Eubacterium rectale
Ruminococcus bromii
Bifidobacterium adolescentis
Alistipes putredinis
Gemmiger formicilis
Bifidobacterium pseudocatenulatum
Phocaeicola plebeius
Phocaeicola dorei
Streptococcus salivarius
Bifidobacterium longum
Anaerostipes hadrus
Mediterraneibacter faecis
Fusicatenibacter saccharivorans
Akkermansia muciniphila
Bacteroides stercoris
Dorea longicatena
Phocaeicola coprocola
Ruminococcus bicirculans
Alistipes onderdonkii
Blautia obeum
Simiaoa sunii
Bacteroides eggerthii
Phocaeicola massiliensis
Wujia chipingensis
Megamonas funiformis
Bacteroides cellulosilyticus
Alistipes finegoldii
Sutterella wadsworthensis
Butyrivibrio crossotus
Acidaminococcus fermentans
Enterocloster clostridioformis
Vescimonas coprocola
Clostridiaceae bacterium Marseille Q4149
Phascolarctobacterium faecium
Roseburia sp AF02 12
Pseudoruminococcus massiliensis
Oscillibacter sp MSJ 31
Segatella brunsvicensis
Alistipes dispar
Bacteroides clarus
Candidatus Gastranaerophilales bacterium
Segatella sinensis
Lachnospira sp NSJ 43
Ligilactobacillus salivarius
Klebsiella variicola
Rhizobium oryzihabitans
Veillonella ratti
Veillonellaceae bacterium
Alistipes sp An31A
Achromobacter marplatensis
f
Stool Species Distribution (PDAC)
Universal
Overlapping
Singular
Relative abundance (%)
Vilà−Quintana_2024
Kartal_2022
Meng_2025
Matsukawa_2021
Nagata_2022
Bacteroides uniformis
Phocaeicola vulgatus
Akkermansia muciniphila
Streptococcus salivarius
Segatella copri
Phocaeicola dorei
Faecalibacterium prausnitzii
Escherichia coli
Parabacteroides distasonis
Blautia wexlerae
Bifidobacterium longum
Alistipes onderdonkii
Ruminococcus bromii
Bifidobacterium adolescentis
Bacteroides thetaiotaomicron
Eubacterium rectale
Prevotella buccae
Ruminococcus gnavus
Bacteroides caccae
Ruminococcus torques
Alistipes putredinis
Bacteroides ovatus
Bacteroides stercoris
Veillonella parvula
Streptococcus parasanguinis
Megamonas funiformis
Bifidobacterium breve
Enterobacter ludwigii
Prevotella marseillensis
Segatella sinensis
Bacteroides faecis
Segatella brunsvicensis
Prevotella stercorea
Clostridium sp AM22 11AC
Faecalimonas umbilicata
Megasphaera massiliensis
Sutterella sp AM11 39
Collinsella sp AF08 23
Intestinimonas butyriciproducens
Blautia caecimuris
Klebsiella oxytoca
Clostridiaceae bacterium Marseille Q4149
Tyzzerella nexilis
Megasphaera micronuciformis
Faecalibacterium sp CLA AA H233
Lactobacillus crispatus
Alistipes indistinctus
Dialister pneumosintes
Megasphaera sp NM10
Clostridium sp AF36 4
Butyrivibrio crossotus
Rhizobium oryzihabitans
Lachnoclostridium sp An181
Sutterella massiliensis
Achromobacter marplatensis
g
Top 30 Overlapping Stool Genera (Control)
Average number of species
Bacteroides
Blautia
Clostridium
Alistipes
* Streptococcus (−)
* Ruminococcus (+)
Roseburia
* Enterocloster (−)
Clostridiaceae
* Eubacterium (+)
* Bifidobacterium (+)
Phocaeicola
Dorea
Parabacteroides
Lachnospiraceae
* Veillonella (−)
* Coprococcus (−)
Faecalibacterium
Oscillibacter
Lachnospira
* Oscillospiraceae (+)
* Butyricimonas (−)
Clostridiales
* Actinomyces (−)
Fusicatenibacter
* Mediterraneibacter (+)
* Intestinimonas (−)
* Anaerobutyricum (−)
* Schaalia (−)
* Segatella (−)
h
Top 30 Overlapping Stool Genera (PDAC)
Average number of species
Bacteroides
Blautia
Clostridium
Alistipes
* Streptococcus (+)
* Ruminococcus (−)
Roseburia
* Enterocloster (+)
Clostridiaceae
* Eubacterium (−)
* Bifidobacterium (−)
Phocaeicola
Dorea
Parabacteroides
Lachnospiraceae
* Veillonella (+)
* Coprococcus (+)
Faecalibacterium
Oscillibacter
Lachnospira
* Oscillospiraceae (−)
* Butyricimonas (+)
Clostridiales
* Actinomyces (+)
Fusicatenibacter
* Mediterraneibacter (−)
* Intestinimonas (+)
* Anaerobutyricum (+)
* Schaalia (+)
* Segatella (+)
i
Top 30 Stool Species (Control)
Mean relative abundance (%)
Faecalibacterium prausnitzii
Phocaeicola vulgatus
Ruminococcus bromii
Eubacterium rectale
Bacteroides uniformis
Blautia wexlerae
Segatella copri
Bifidobacterium adolescentis
Bifidobacterium longum
Fusicatenibacter saccharivorans
Phocaeicola dorei
Bifidobacterium pseudocatenulatum
Streptococcus salivarius
Ruminococcus bicirculans
Alistipes putredinis
Phocaeicola plebeius
Gemmiger formicilis
Escherichia coli
Bacteroides stercoris
Blautia massiliensis
Ruminococcus gnavus
Megamonas funiformis
Anaerostipes hadrus
Phocaeicola coprocola
Dorea longicatena
Mediterraneibacter faecis
Ruminococcus torques
Parabacteroides distasonis
Lachnospira eligens
Roseburia faecis
j
Top 30 Stool Species (PDAC)
Mean relative abundance (%)
Bacteroides uniformis
Phocaeicola vulgatus
Akkermansia muciniphila
Streptococcus salivarius
Phocaeicola dorei
Faecalibacterium prausnitzii
Segatella copri
Blautia wexlerae
Parabacteroides distasonis
Escherichia coli
Ruminococcus bromii
Alistipes onderdonkii
Eubacterium rectale
Bifidobacterium longum
Bifidobacterium adolescentis
Bacteroides thetaiotaomicron
Prevotella buccae
Bacteroides stercoris
Ruminococcus torques
Bacteroides ovatus
Ruminococcus gnavus
Bacteroides caccae
Bacteroides fragilis
Alistipes putredinis
Roseburia faecis
Streptococcus parasanguinis
Megamonas funiformis
Lachnospira eligens
Anaerostipes hadrus
Fusicatenibacter saccharivorans